

4d) AIM

To write a C program to perform Round Robin Scheduling

Algorithm

Step 1: Start

Step 2: Declare a structure process with attributes pid, bt, wt, tat and tbt

Step 3: Declare variables i, n, x=0, y=0, tq, avgw=0, avgt=0 and p of size 10.

Step 4: Read the number of process, n

Step 5: Read the time quantum, tq.

Step 6: Read the process id, burst time for each process and store it in p.

Step 7: Read the values of bt into tbt and store it in p.

Step 8: let i=0

Step 9: while true,

Step 9.1: if p[i].bt is not 0,

Step 9.1.1: do if p[i].bt > time quantum,

let x = x + time quantum and p[i].bt = p[i].bt - time quantum.

Step 9.1.2: else

add p[i].bt with x

let p[i].bt = 0 and increment the value of y.

let p[i].tat = x and p[i].wt = x - p[i].tbt

break if y = n.

Step 9.2: if i = n-1, then let i = 0

Step 9.3: else, increment i

Step 10: Calculate the average waiting time and average turnaround time.

Step 11: Display process details with average waiting time and average turnaround time.

Step 12: Stop.

Result: The program was executed successfully and output was verified.